# Lecture 6: Excessive Volatility in Prices and Returns

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# Intro

# Do Stock Prices Move Too Much to be Justified by Subsequent Changes in Dividends?

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- Equity prices move around a lot... can dividend expectation changes justify this?

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- $D_{t+k}$ = dividend at time $t+k$
- $r$ = constant discount rate
- With constant $r$, can this model generate the same volatility in prices as seen in the data?
- Price changes **must** be driven by news about future dividends!

# Realized vs Forecasted Dividends

- The "efficient market model" implies:

$$P_t = \mathbb{E}_t \left[ \sum_{k=1}^{\infty} \frac{D_{t+k}}{(1+r)^k} \right] = \sum_{k=1}^{\infty} \gamma^{k+1} \mathbb{E}_t [D_{t+k}], \qquad \gamma \equiv \frac{1}{1+r}$$

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- Realized dividends are the sum of expected dividends and an "innovation" (forecast error):

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- The realized series cannot be smoother than its projection:

$$\sigma^2(D_{t+k}) = \sigma^2(\mathbb{E}_t[D_{t+k}] + \varepsilon_{t+k}) = \sigma^2(\mathbb{E}_t[D_{t+k}]) + \sigma^2(\varepsilon_{t+k}) \geq \sigma^2(\mathbb{E}_t[D_{t+k}])$$

## Shiller's Key Insight

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- Let $\Pi_t$ be the GDP deflator and $A \equiv 1 + g$, where $g$ a growth rate for real dividends
- Shiller deflated the index prices and dividends by inflation and the growth rate:

$$p_t \equiv \frac{P_t}{\Pi_t A^{t-T}}, \qquad d_t \equiv \frac{D_t}{\Pi_t A^{t+1-T}}$$

where $T = 1978$ is the last year of the sample and $g$ is estimated as the constant log-growth rate of real dividends.

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- Similarly: $p_t^* = \sum_{k=1}^{\infty} \tilde{\gamma}^{k+1} d_{t+k}$

## Shiller's Key Insight

- Crucially, we have the relationship:

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- Dividends after 1978 are unknown at time $t$
- Shiller assumed that they would grow at rate $g$ forever
- This assumption likely understates $\sigma^2(p_t^*)$
- $p_{1978}^*$ is computed with "fake dividends" and then you use the recursion backward:

$$p_t^* = \tilde{\gamma}(d_{t+1} + p_{t+1}^*)$$

- The discount rate $r$ is such that $r = \frac{\mathbb{E}[d]}{\mathbb{E}[p]}$ (why was deflating necessary for this step?)

# The Main Plot

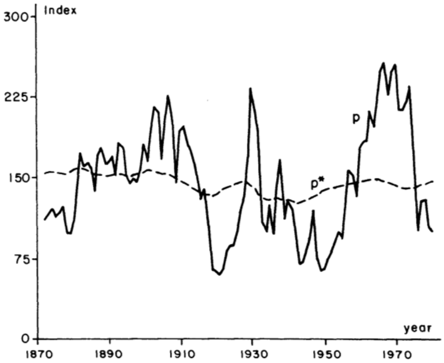



FIGURE 1

*Note*: Real Standard and Poor's Composite Stock Price Index (solid line $p$) and *ex post* rational price (dotted line $p^*$), 1871–1979, both detrended by dividing a long-run exponential growth factor. The variable $p^*$ is the present value of actual subsequent real detrended dividends, subject to an assumption about the present value in 1979 of dividends thereafter. Data are from Data Set 1, Appendix.

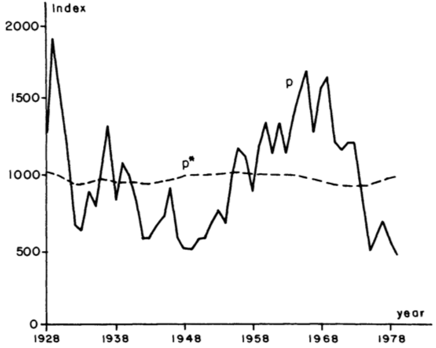



FIGURE 2

*Note*: Real modified Dow Jones Industrial Average (solid line $p$) and *ex post* rational price (dotted line $p^*$), 1928-1979, both detrended by dividing by a long-run exponential growth factor. The variable $p^*$ is the present value of actual subsequent real detrended dividends, subject to an assumption about the present value in 1979 of dividends thereafter. Data are from Data Set 2, Appendix.

# Main Numerical Result

- Strange things: inequalities upside down!
- You see many rows because he derived many bounds with different assumptions
- The data is rejecting theory by an order of magnitude
- $50 > 8$ and $355.9 > 26.8$

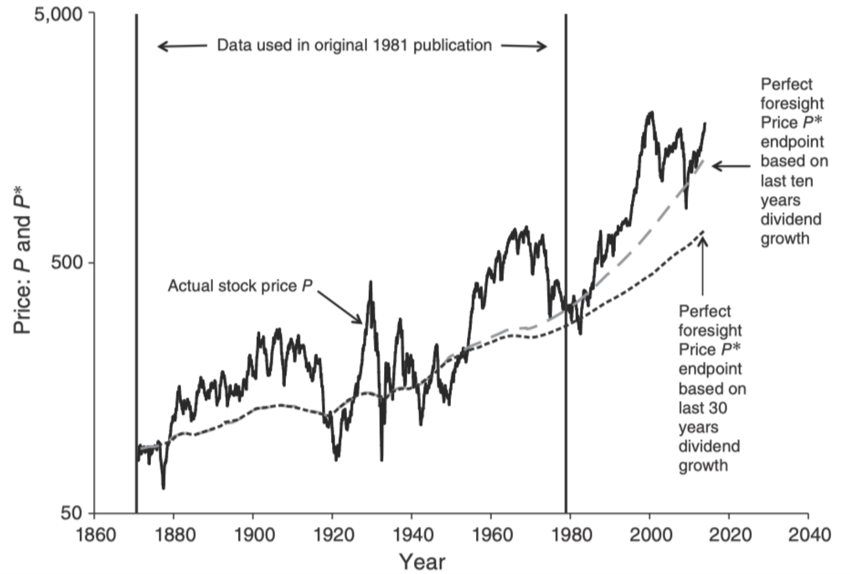
**Historical evidence**:

- During the Great Depression, (real) dividends were below average only on 1933-1935, and 1939
- But returns were pretty bad!

TABLE 2—SAMPLE STATISTICS FOR PRICE AND DIVIDEND SERIES

| | | Data Set 1: Standard and Poor's | Data Set 2: Modified Dow Industrial |
|---|---|---|---|
| | Sample Period: | 1871–1979 | 1928–1979 |
| 1) | $E(p)$ | 145.5 | 982.6 |
| | $E(d)$ | 6.989 | 44.76 |
| 2) | $\bar{r}$ | .0480 | 0.456 |
| | $\bar{r}_2$ | .0984 | .0932 |
| 3) | $b = ln\lambda$ | .0148 | .0188 |
| | $\hat{\sigma}(b)$ | (.0011) | (1.0035) |
| 4) | $cor(p, p^*)$ | .3918 | .1626 |
| | $\sigma(d)$ | 1.481 | 9.828 |
| Elements of Inequalities: | | | |
| Inequality (1) | | | |
| 5) | $\sigma(p)$ | 50.12 | 355.9 |
| 6) | $\sigma(p^*)$ | 8.968 | 26.80 |
| Inequality (11) | | | |
| 7) | $\sigma(\Delta p + d_{-1} - \bar{r}p_{-1})$ | 25.57 | 242.1 |
| | $min(\sigma)$ | 23.01 | 209.0 |
| 8) | $\sigma(d)/\sqrt{\bar{r}_2}$ | 4.721 | 32.20 |
| Inequality (13) | | | |
| 9) | $\sigma(\Delta p)$ | 25.24 | 239.5 |
| | $min(\sigma)$ | 22.71 | 206.4 |
| 10) | $\sigma(d)/\sqrt{2\bar{r}}$ | 4.777 | 32.56 |

# Newer Data


5,000
500
50
Price: P and P*
Data used in original 1981 publication
Actual stock price P
Perfect foresight Price P* endpoint based on last ten years dividend growth
Perfect foresight Price P* endpoint based on last 30 years dividend growth
1860
1880
1900
1920
1940
1960
1980
2000
2020
2040
Year

## What if Discount Rates Vary?

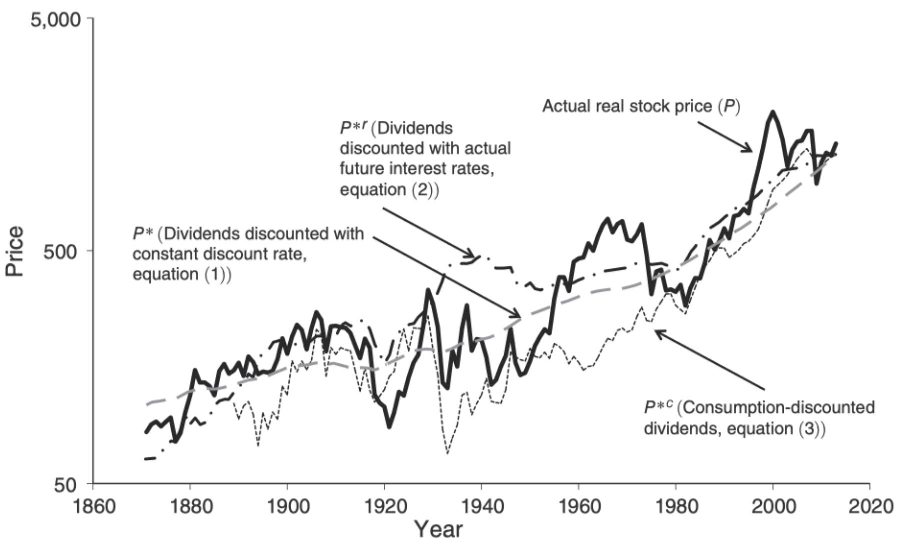
- We can always write:

$$P_t = \mathbb{E}_t \left[ \sum_{k=1}^{\infty} D_{t+k} \prod_{j=1}^{k} \frac{1}{1 + r_{t+j-1}} \right]$$

- If we impose no discipline, this will always be true for some series of returns
- Shiller (1981) used 6-month prime commercial paper + a constant
- Shiller (2014) used our good and old friend:

$$r_{t+1} = -\log(\beta) + \delta \cdot \Delta c_{t+1}$$

# Newer Data and Time-Varying Interest Rates



# Where The Literature Took This

**Takeaway message**:

- Stock prices move too much to be justified by subsequent changes in dividends
- The natural way to reconcile this is allowing for rich time-variation in discount rates

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**Two Paths Forward**:

1. **Return decompositions**: how much of price movements is due to changes in dividend expectations vs expected returns? (Campbell and Shiller 1988; Campbell 1991; Cochrane 2011)
2. **Behavioral finance and limits to arbitrage**: prices deviate from fundamentals due to investor psychology and limits to arbitrage (De Long et al. 1990; Shleifer and Vishny 1997; Shiller 2014; Barberis and Thaler 2003; Barberis 2018)

## Behavioral Finance

- Excess volatility could reflect investor psychology, not rational risk premia
- Noise traders and limits to arbitrage (De Long et al. 1990; Shleifer and Vishny 1997)
- Shiller (2014): irrational exuberance and social dynamics drive speculative prices

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- Shiller (2014): irrational exuberance and social dynamics drive speculative prices
- Surveys: Barberis and Thaler (2003) and Barberis (2018) (extrapolation, overconfidence, prospect theory)
- We will not pursue this path, but it is an important and active research program

**Questions?**

# Campbell–Shiller Log-Linearization (Setup)

- Exact log return identity:

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- Rewrite the nonlinear term:

$$\log(P_{t+1} + D_t) = p_{t+1} + \log(1 + \exp(d_t - p_{t+1}))$$

- Define the log dividend–price ratio:

$$\delta_{t+1} \equiv d_t - p_{t+1}$$

- Dividend $D_t$ is paid at time $t + 1$ in this notation (keeping it close to the paper)

# Campbell–Shiller Log-Linearization (Taylor Step)

- Let $g(\delta) \equiv \log(1 + e^{\delta})$ and expand around $\bar{\delta}$:

$$g(\delta_{t+1}) \approx g(\bar{\delta}) + g'(\bar{\delta})(\delta_{t+1} - \bar{\delta})$$

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- Derivative and the key constant $\rho$:

$$g'(\delta) = \frac{e^{\delta}}{1 + e^{\delta}}, \qquad 1 - \rho \equiv g'(\bar{\delta}), \qquad \rho = \frac{1}{1 + e^{\bar{\delta}}}$$

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- Collect constants into $k$:

$$\log(P_{t+1} + D_t) \approx k + \rho\, p_{t+1} + (1 - \rho)\, d_t$$

## Campbell–Shiller Log-Linear Return + Forward Solution

- Substitute into the return identity:

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- Rearranged recursion and forward solution:

$$\delta_t = (h_t - \Delta d_t - k) + \rho\, \delta_{t+1} \;\Rightarrow\; \boxed{\delta_t = \sum_{j=0}^{\infty} \rho^j (h_{t+j} - \Delta d_{t+j}) - \frac{k}{1-\rho}}$$

- This equation has no economic content: it is an (approximate) **identity**. Intuition?

# How to Add Empirical Content

- Notice that we can take expectations conditional on information at time $t$:

$$\delta_t = \mathbb{E}_t\left[\sum_{j=0}^{\infty}\rho^j(h_{t+j} - \Delta d_{t+j})\right] - \frac{k}{1-\rho} = \sum_{j=0}^{\infty}\rho^j(\mathbb{E}_t[h_{t+j}] - \mathbb{E}_t[\Delta d_{t+j}]) - \frac{k}{1-\rho}$$

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- Example: suppose that for some $r_{t+j}$ and some $c$ we have $\mathbb{E}_t[h_{t+j}] = \mathbb{E}_t[r_{t+j}] + c$
- We can write:

$$\delta_t = \sum_{j=0}^{\infty} \rho^j (\mathbb{E}_t[r_{t+j}] - \mathbb{E}_t[\Delta d_{t+j}]) + \frac{c-k}{1-\rho}$$

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- This will become testable once we specify a model for returns and dividends
- Example: give me $\tilde{r}_{t,j} \equiv \mathbb{E}_t[r_{t+j}]$ and $\tilde{\Delta} d_{t,j} \equiv \mathbb{E}_t[\Delta d_{t+j}]$ and generate $\tilde{\delta}_t$
- We should have $\delta_t \approx \tilde{\delta}_t$

# Questions?

# Campbell and Shiller Meet the Data

## Setup

- Assume that market participants use state variables $y_t$ to do forecasts
- $y_t$ evolves linearly (think about an MA($\infty$) process)
- The econometrician observes (with abbuse of notation) $x_t \equiv (\delta_t, h_{t-1}, \Delta d_{t-1})^\top \subset y_t$

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- Assume that $x_t$ (already demeaned) follows a VAR($p$):

$$x_t = C_1 x_{t-1} + ... + C_p x_{t-p} + u_t$$

where all variables are demeaned and $u_t$ is a white noise process each $C_i$ is a $3 \times 3$ matrix

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- Rewrite the VAR in companion form:

$$z_t \equiv \begin{bmatrix} x_t \\ x_{t-1} \\ \vdots \\ x_{t-p+1} \end{bmatrix} = \begin{bmatrix} C_1 & C_2 & ... & C_p \\ I & 0 & ... & 0 \\ 0 & I & ... & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & ... & 0 \end{bmatrix} \begin{bmatrix} x_{t-1} \\ x_{t-2} \\ \vdots \\ x_{t-p} \end{bmatrix} + \begin{bmatrix} u_t \\ 0 \\ \vdots \\ 0 \end{bmatrix} = A z_{t-1} + v_t$$

# The VAR Approach

- The companion form implies: $\mathbb{E}_t[z_{t+j}] = A^j z_t$
- The VAR is a forecasting machine to generate discount rate and dividend forecasts
- Given $A$, we can compute:

$$\tilde{\delta}_t = \sum_{j=0}^{\infty} \rho^j \big(\mathbb{E}[h_{t+j}|\{x_t, x_{t-1}, ...\}] - \mathbb{E}[\Delta d_{t+j}|\{x_t, x_{t-1}, ...\}]\big)$$

where $\tilde{r}_{t,j}$ and $\tilde{\Delta} d_{t,j}$ are the appropriate elements of $\mathbb{E}_t[z_{t+j}]$

# There is hope in the approximation!

**Table 2**
**Summary statistics for stock market data**

| Statistic | Data set and sample period: Cowles/S&P, 1871–1986 | Cowles/S&P, 1926–1986 | NYSE, 1926–1986 | Correlation of Cowles/S&P and NYSE, 1926–1986 |
|---|---|---|---|---|
| $\Delta p_t$: | | | | |
| Mean | 0.032 | 0.044 | 0.042 | 0.972 |
| Standard deviation | 0.178 | 0.200 | 0.208 | |
| $\Delta d_t$: | | | | |
| Mean | 0.030 | 0.041 | 0.040 | 0.958 |
| Standard deviation | 0.132 | 0.131 | 0.134 | |
| $\delta_t$: | | | | |
| Mean | −3.053 | −3.121 | −3.143 | 0.985 |
| Standard deviation | 0.277 | 0.294 | 0.290 | |

All variables in this table are nominal and measured annually. $p_t$ is the log stock price, $d_t$ is the log dividend, and $\delta_t$ is the log dividend-price ratio $d_{t-1} - p_t$.

# AI-Generated Summary

# Readings and References i


Barberis, Nicholas. 2018. "Psychology-Based Models of Asset Prices and Trading Volume." In *Handbook of Behavioral Economics: Applications and Foundations*, 1:79–175. Elsevier. https://doi.org/10.1016/bs.hesbe.2018.07.001.

Barberis, Nicholas, and Richard Thaler. 2003. "A Survey of Behavioral Finance." In *Handbook of the Economics of Finance*, 1:1053–1128. Elsevier. https://doi.org/10.1016/S1574-0102(03)01027-6.

Campbell, John Y. 1991. "A Variance Decomposition for Stock Returns." *The Economic Journal* 101 (405): 157–79. https://doi.org/10.2307/2233809.

Campbell, John Y., and Robert J. Shiller. 1988. "The Dividend-Price Ratio and Expectations of Future Dividends and Discount Factors." *The Review of Financial Studies* 1 (3): 195–228. https://doi.org/10.1093/rfs/1.3.195.

Cochrane, John H. 2011. "Presidential Address: Discount Rates." *The Journal of Finance* 66 (4): 1047–1108. https://doi.org/10.1111/j.1540-6261.2011.01671.x.

De Long, J. Bradford, Andrei Shleifer, Lawrence H. Summers, and Robert J. Waldmann. 1990. "Noise Trader Risk in Financial Markets." *Journal of Political Economy* 98 (4): 703–38. https://doi.org/10.1086/261703.

# Readings and References ii


Shiller, Robert J. 1981. "Do Stock Prices Move Too Much to Be Justified by Subsequent Changes in Dividends?" *The American Economic Review* 71 (3): 421–36. http://www.jstor.org/stable/1802789.

———. 2014. "Speculative Asset Prices." *American Economic Review* 104 (6): 1486–1517. https://doi.org/10.1257/aer.104.6.1486.

Shleifer, Andrei, and Robert W. Vishny. 1997. "The Limits of Arbitrage." *The Journal of Finance* 52 (1): 35–55. https://doi.org/10.1111/j.1540-6261.1997.tb03807.x.